

Julia Goh

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Experienced AI Engineer (Applications, Deployment and Software/Full-Stack). MEng Computer Science graduate at University College London. Fluent in English, Chinese and Malay. Open to senior/lead roles in ML/AI Engineering, Software Engineering and Full Stack. Acquired Microsoft certifications of AI-900, AI-102 and DP-900.

Education

University College London (UCL) | MEng Computer Science | September '20 – June '24

- **Grade**: First Class Honours
- Minor in Intelligent Systems
- Treasurer of UCL Portuguese Speaking Society 22/23
- Computer Graphics: Ray Tracing, Rasterization and Path Tracing with **C++**.
- Robotics: Forward and Inverse Kinematics, Path Planning and Robot Dynamics with ROS.
- Football Match Prediction: Regression, Classification, Tree and Forest, and SVM with **Python**.

Experiences

Unify | Founding AI Engineer (Full Stack) | April '25 – Present

- Agentic AI Systems: Developed LLM agentic systems, leveraging harness and context engineering for enabling tools, loops and various task executions. Introduced knowledge and skills modules as memory for agents to reference on task execution. Also involved RAG by wiring up relevant integrations and searches (including web) to further widen agent's awareness where relevant.
- AI Assistant Communications: Enabled the omni-channel communications infrastructure powering the AI assistant — voice calls, multi-channel messaging (Slack, Microsoft Teams, Discord, WhatsApp), emails, and in-app chat/calls — integrating Twilio, LiveKit, platform bot frameworks and third-party APIs.
- Live Meeting Attendance: Built the assistant's ability to join Google Meet and Microsoft Teams calls as a real participant, with two-way screenshare, in-meeting chat and auto-leave — via a Recall.ai bridge.
- Voice Identity & Call Recording: Built speaker diarization with voice-fingerprint labelling, splitting co-located speakers sharing a single microphone through per-utterance embedding clustering, alongside call recording/egress and speech-aligned transcripts with in-app playback.
- AI Desktop & Browser Control: Built a vision-based (vLLM) computer-use agent that automates tasks across remote Ubuntu, Windows and macOS desktops, with CLI-installable clients for connecting a user's own machine.
- Organisation Support: Built multi-tenant organisation features on top of personal workspaces — member management and invites, role-based access control with custom roles, ownership transfer, and project/resource sharing.
- **Tech Stack: Python, TypeScript, Node.js, React.js, Next.js, FastAPI, PostgreSQL, Docker, Kubernetes, GCP, LiveKit, Playwright, Git**

Petronas | Machine Learning Engineer (Intermediate) | September '24 – March '25

- End-to-end model development and deployment, such as data analysis, model training, hyperparameters tuning, model selection and cloud deployment on Azure.
- Worked with linear models, trees, random forest, XG boost and light gradient boosting machine (LGBM).
- PrAnCorr: *ML.* Built pipeline external corrosion depth & length prediction models of >90% accuracy.
- PrAnSpan: *ML.* Built subsea pipeline height & length prediction models of >70% accuracy.
- HIP: *Computer Vision.* Worked on working person detection through object detection with YOLO, pose estimation and vLLM with GPT4o. Involved prompt engineering for better inference performance.
- **Tech Stack**: **Python, MLFlow, Docker, SQL, Azure, SonarCloud, DevOps, Git**

Unify | Machine Learning Engineer (Full Stack) | June '22 – February '24

- Ivy ML Models: *Applied Lead*. Led the team to expand ML model collection from scratch to 15+ within 4 months, including computer vision and NLP base models for business showcase.
- Ivy Transpiler: Introduced the native compilation feature for optimising performance of ML models.
- Model Hub: *Full stack*. Responsible in building client applications and API key authentication.
- Console Application: *Cloud*. Involved in v0 development and API gateway authorisation.
- Ivy: *Software Engineering*. Unified PyTorch, Tensorflow, JAX and NumPy functions, making them easily accessible through a single library. All codes are also readily compatible with the native libraries.
- **Tech Stack**: **Python, C++, CUDA, AMD RocM, Node.js, React.js, Next.js, TypeScript, Docker, GCP**

Research

UCL | Generative AI in Music | September '23 – April '24

- Review. Studied existing tech related to image and music generations, majority being text-conditioned.
- Paper. Developed POCs for Diffusion based Music-conditioned Image Generative AI and Transformer based Image-conditioned Music Generative AI.
- **Tech Stack: Python, Pillow, NumPy, SciPy, PyTorch, Transformers, Diffusers, Matplotlib, Git**

UCL | Network Analysis of Programming Languages | November '23 – January '24

- Applied PageRank and HITS for analysing key driver in the evolution network.
- Analysed the community structure of the network with modularity.
- **Tech Stack: Python, NetworkX, NumPy, Matplotlib, Git**

UCL | Multi-Agent Reinforcement Learning in Portfolio Management | January '23 – April '23

- Paper. Explored the effect of different RL algorithms in competitive vs. cooperative MARL settings.
- Algorithms: Advantaged Actor-Critic (A2C), Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimisation (PPO)
- **Tech Stack: Python, OpenAI Gym, Stable Baselines 3, Pandas, Git**